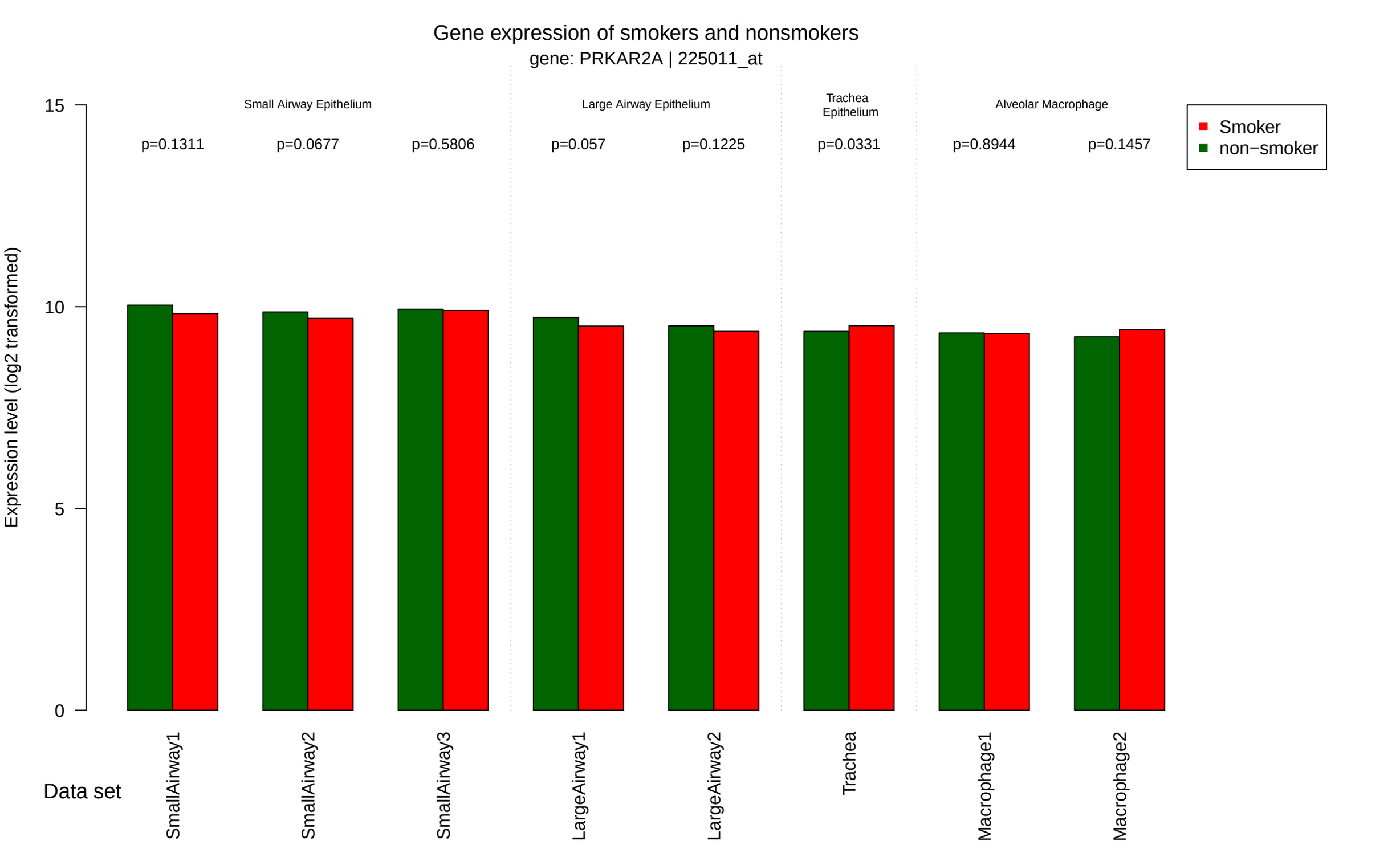
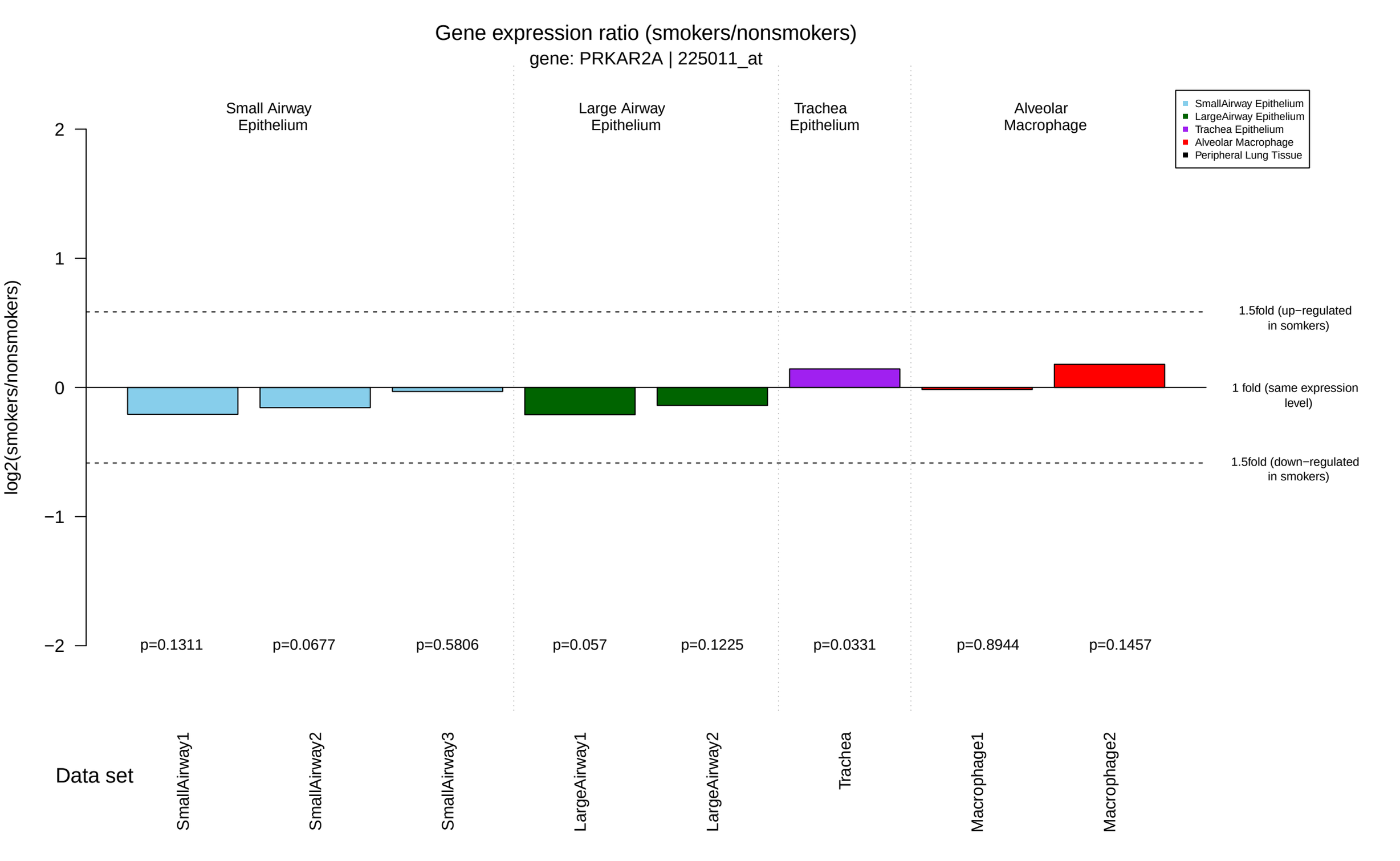
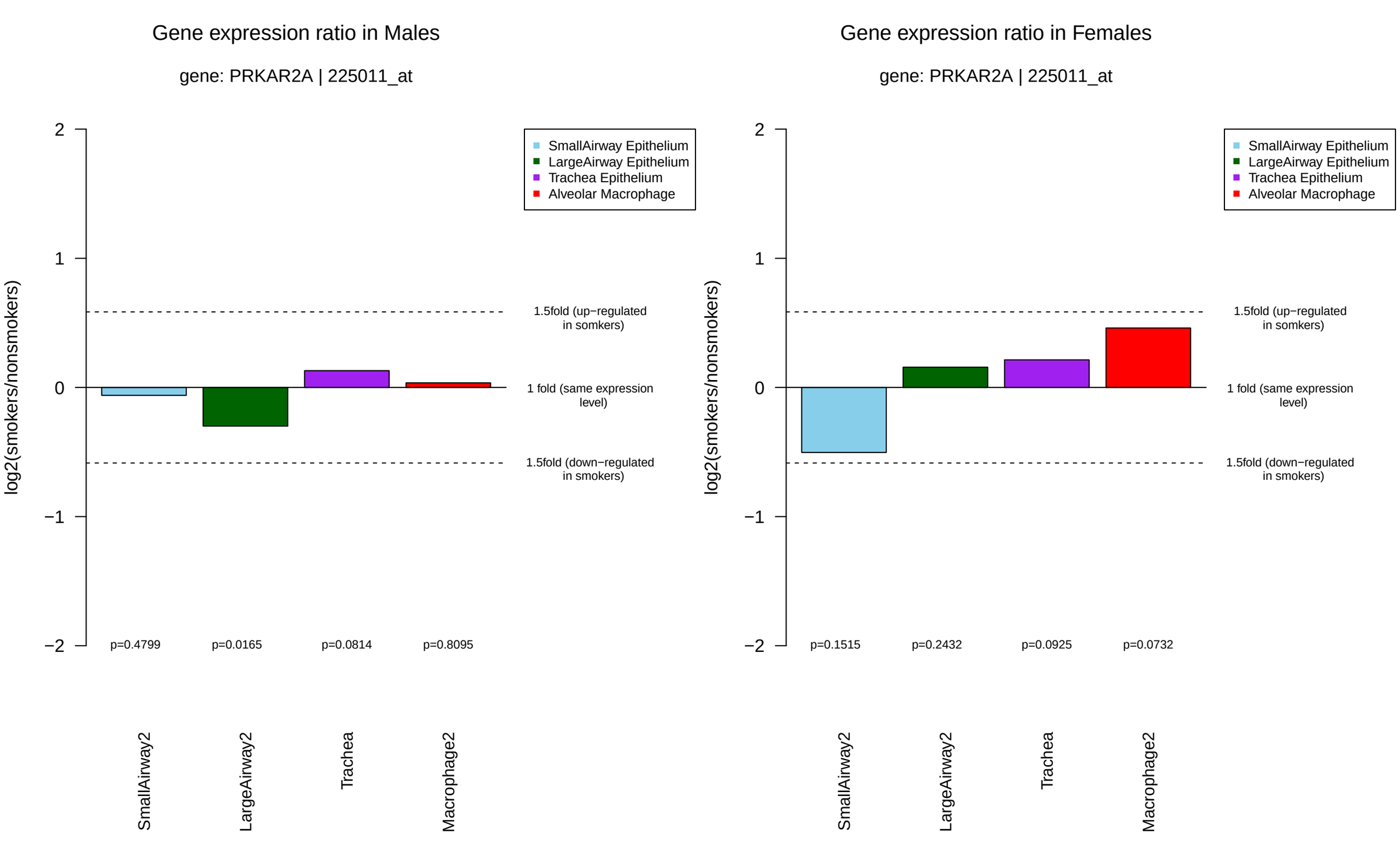
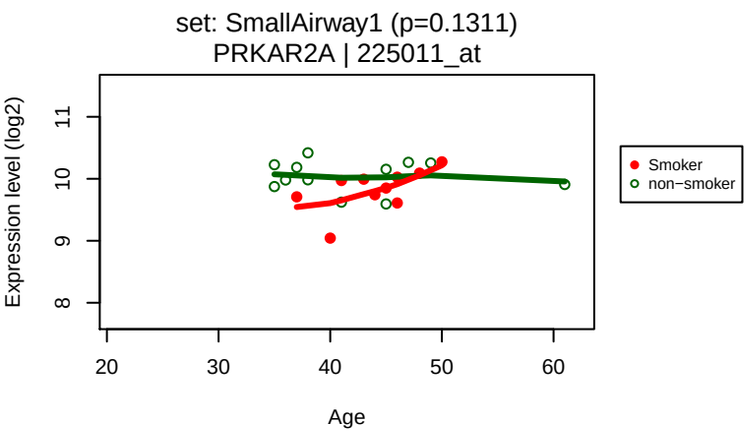
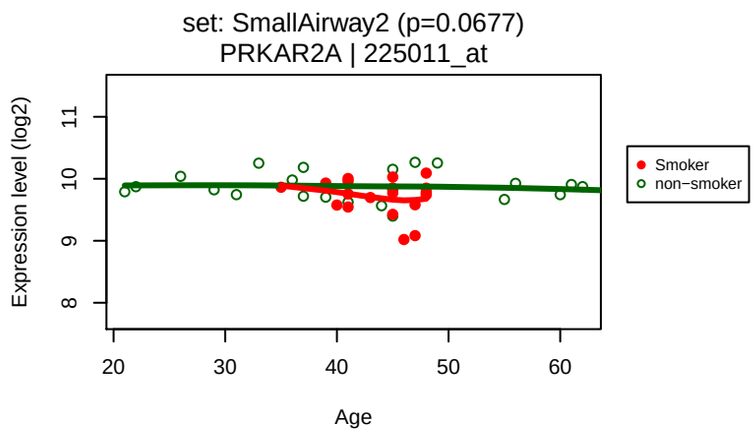
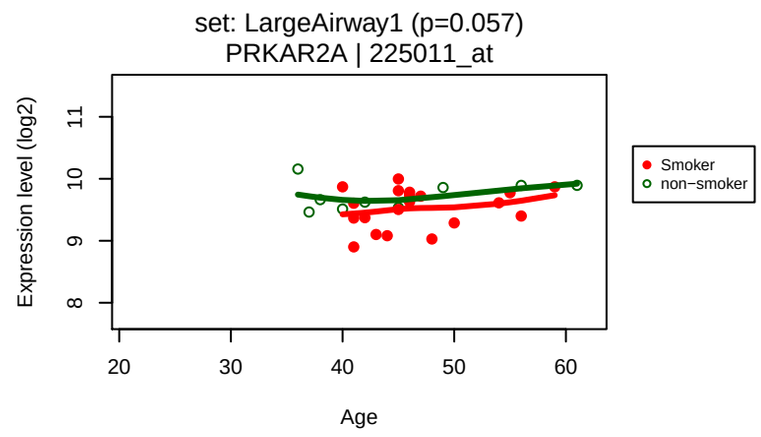

Gene expression of smokers and nonsmokers
gene: PRKAR2A | 225011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1311
p=0.0677
p=0.5806
p=0.057
p=0.1225
p=0.0331
p=0.8944
p=0.1457
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

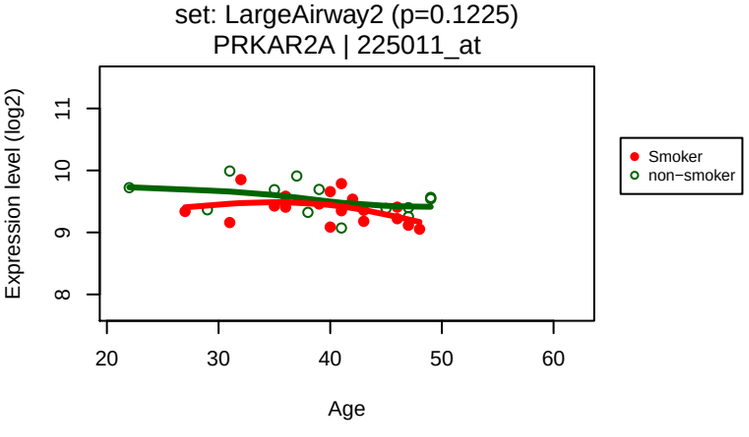

Gene expression ratio (smokers/nonsmokers)
gene: PRKAR2A | 225011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1311
p=0.0677
p=0.5806
p=0.057
p=0.1225
p=0.0331
p=0.8944
p=0.1457
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

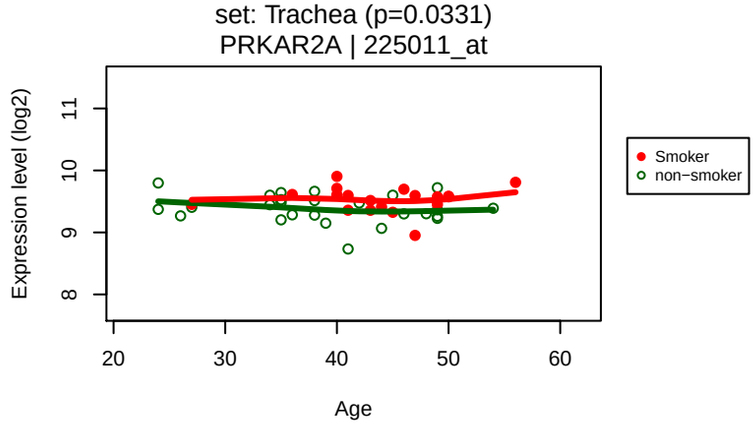

Gene expression ratio in Males
gene: PRKAR2A | 225011_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4799
p=0.0165
p=0.0814
p=0.8095
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAR2A | 225011_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1515
p=0.2432
p=0.0925
p=0.0732
SmallAirway2
LargeAirway2
Trachea
Macrophage2

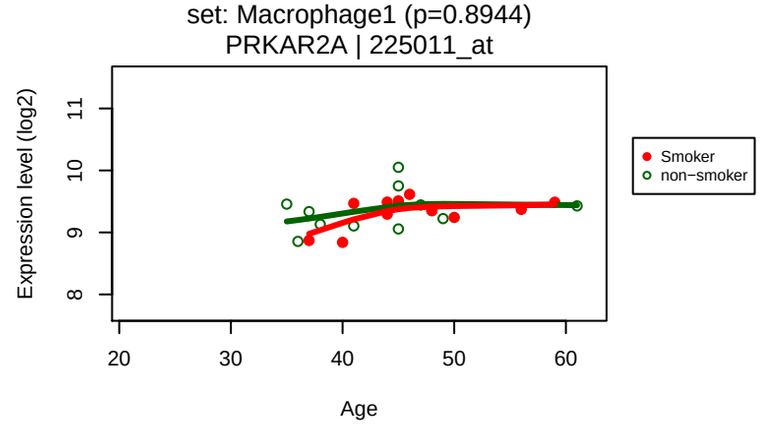

set: SmallAirway1 (p=0.1311)
PRKAR2A | 225011_at
Expression level (log2)
Age
Smoker
non−smoker

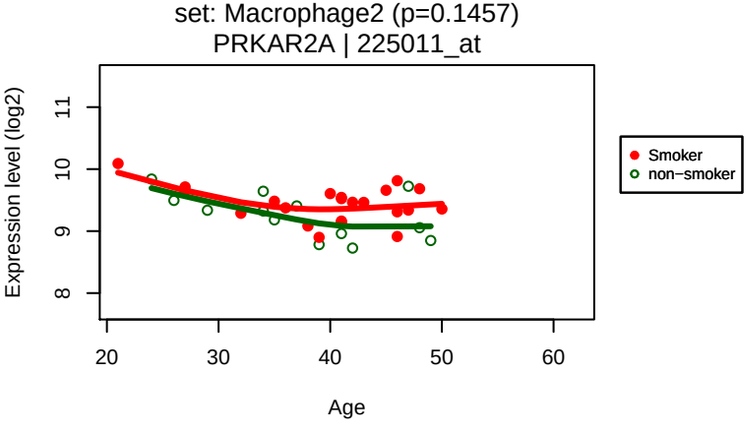

set: SmallAirway2 (p=0.0677)
PRKAR2A | 225011_at
Expression level (log2)
Age
Smoker
non−smoker

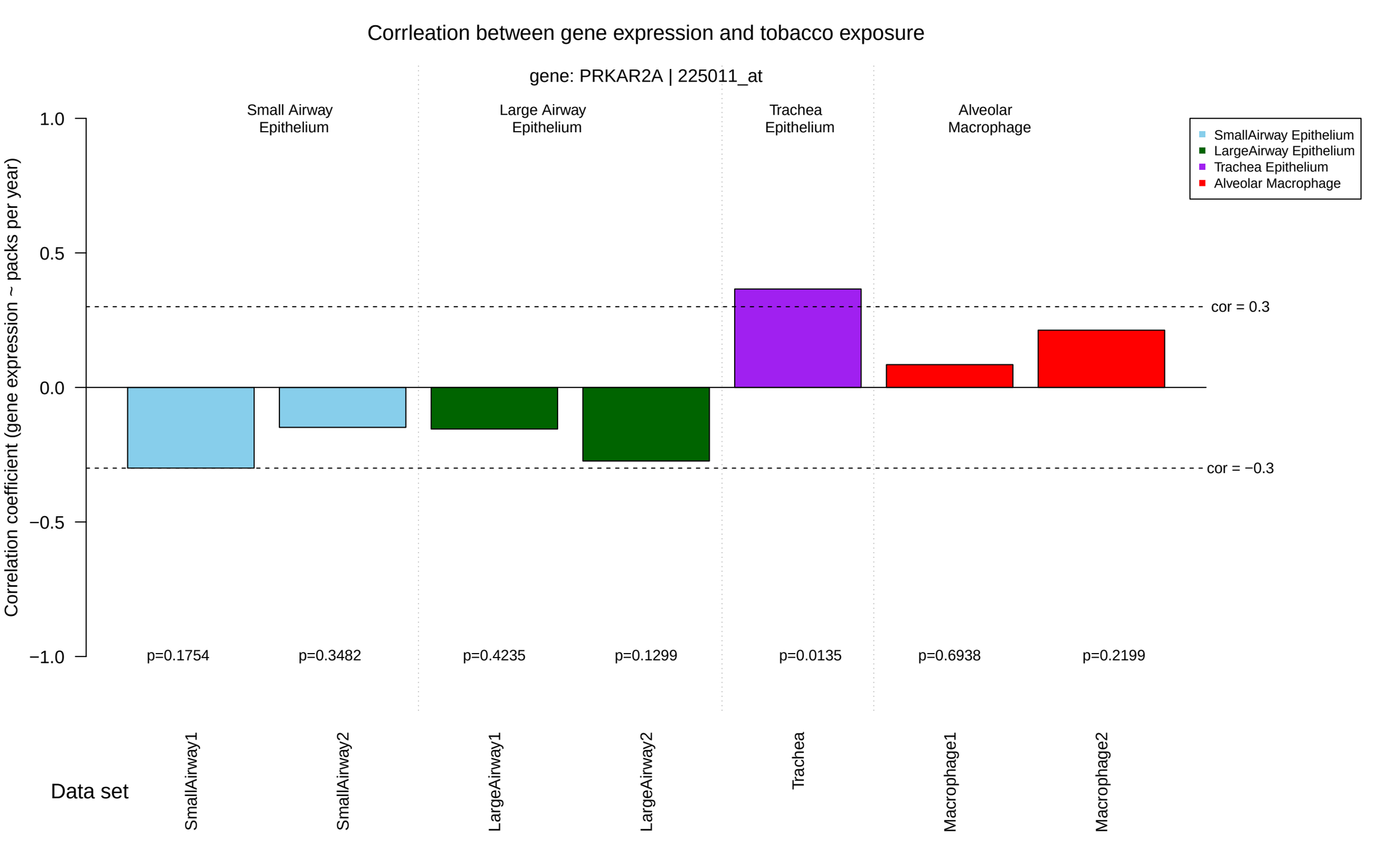

set: LargeAirway1 (p=0.057)
PRKAR2A | 225011_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1225)
PRKAR2A | 225011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0331)
PRKAR2A | 225011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8944)
PRKAR2A | 225011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1457)
PRKAR2A | 225011_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAR2A | 225011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1754
p=0.3482
p=0.4235
p=0.1299
p=0.0135
p=0.6938
p=0.2199
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2